

```
%%writefile linear_regression.py
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
import joblib

data = {
    'YearsExperience': [1,2,3,4,5,6,7,8,9,10],
    'Salary': [30000,35000,40000,45000,50000,55000,60000,65000,70000,75000]
}

df = pd.DataFrame(data)

X = df[['YearsExperience']]
y = df['Salary']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

joblib.dump(model, 'linear_model.pkl')

print("Model trained and saved successfully!")
```

```
!pip install streamlit
```

Collecting streamlit

```
  Downloading streamlit-1.56.0-py3-none-any.whl.metadata (9.8 kB)
Requirement already satisfied: altair!=5.4.0,!5.4.1,<7,>=4.0 in /usr/loc
Requirement already satisfied: blinker<2,>=1.5.0 in /usr/local/lib/python
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Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.1
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.12
Requirement already satisfied: pandas<4,>=1.4.0 in /usr/local/lib/python3
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.2-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=3.20 in /usr/local/lib/python
Requirement already satisfied: pyarrow>=7.0 in /usr/local/lib/python3.12/
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python
Requirement already satisfied: tenacity<10,>=8.1.0 in /usr/local/lib/pyth
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.
Requirement already satisfied: tornado!=6.5.0,<7,>=6.0.3 in /usr/local/li
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/pytho
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Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3
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Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.1
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Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist
Downloading streamlit-1.56.0-py3-none-any.whl (9.1 MB)
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Downloading pydeck-0.9.2-py2.py3-none-any.whl (11.3 MB)
_____ 11.3/11.3 MB 80.2 MB/s
Installing collected packages: pydeck, streamlit
Successfully installed pydeck-0.9.2 streamlit-1.56.0
```

```
!pip install streamlit pyngrok
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Requirement already satisfied: streamlit in /usr/local/lib/python3.12/dis
Collecting pyngrok
  Downloading pyngrok-8.1.1-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: altair!=5.4.0,!5.4.1,<7,>=4.0 in /usr/loc
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Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist
Downloading pyngrok-8.1.1-py3-none-any.whl (25 kB)
Installing collected packages: pyngrok
Successfully installed pyngrok-8.1.1
```

```
!ngrok config add-authtoken 3CyKDQ2Rb0L2b3nORGPbwBmSihw_2ngTBuFEFbw33RfKzP1TL
```

```
Authtoken saved to configuration file: /root/.config/ngrok/ngrok.yml
```

```
!ls
```

```
linear_model.pkl  sample_data
```

```
%%writefile app.py
import streamlit as st
import joblib

# Load model
model = joblib.load('linear_model.pkl')

st.title("Salary Prediction App")

years = st.number_input("Years of Experience", min_value=0.0)

if st.button("Predict"):
    result = model.predict([[years]])
    st.success(f"Predicted Salary: ₹ {result[0]:,.2f}")
```

```
Writing app.py
```

```
!ls
```

```
app.py  linear_model.pkl  sample_data
```

```
from pyngrok import ngrok

public_url = ngrok.connect(8501)
print(public_url)
```

```
!streamlit run app.py &
```

NgrokTunnel: "<https://relapsing-scorer-slit.ngrok-free.dev>" -> "<http://localhost:8501>"

Collecting usage statistics. To deactivate, set `browser.gatherUsageStats`

You can now view your Streamlit app in your browser.

Local URL: <http://localhost:8501>

Network URL: <http://172.28.0.12:8501>

External URL: <http://34.138.177.14:8501>

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739:  
warnings.warn(  

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